

Inspectable Symbolic Compression for Long-Context Reasoning

A Mixed-Results Study

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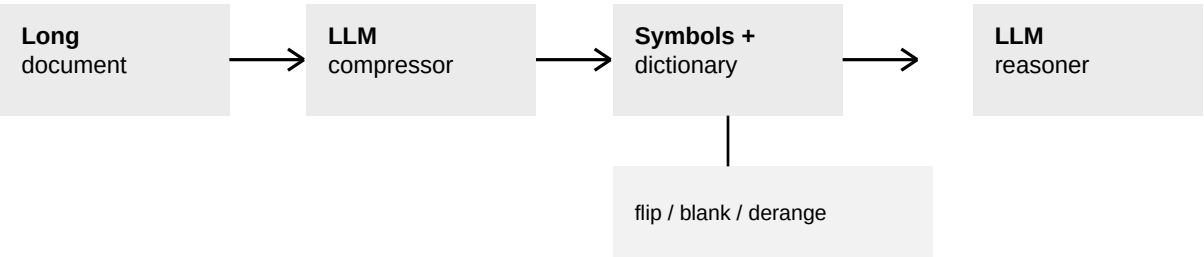
Abstract

We study Inspectable Symbolic Compression (ISM), a prompt-only method that rewrites long documents as discrete symbols, dictionary definitions, and symbolic relations. The result is mixed. ISM structure is functionally used: flipping dictionary conclusions and removing symbolic structure significantly reduce accuracy. However, natural-language model summaries are significantly more accurate and more token-efficient under the same fixed budgets, and they dominate the reuse cost-accuracy frontier. We therefore do not claim that prompt-only ISM is a better compressor. The study instead identifies a boundary: symbolic compression generated by an LLM and read by an LLM remains in the same class as text summarization, pointing toward future executable semantic IRs.

Summary of Findings

RQ	Result	Interpretation
RQ1	ISM dictionary and symbolic structure are	Positive
RQ3	Model Summary beats ISM at all budgets.	Negative
RQ4	Reuse helps caches, but summary dominates	Negative
RQ2	Unseen-label swap remains unrun.	Future work

Figure 1: Prompt-only ISM pipeline.



Method and Setup

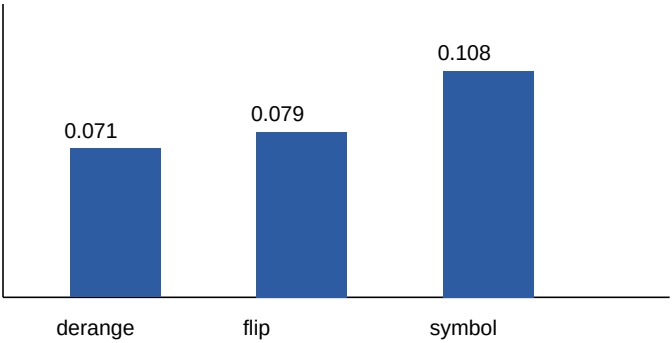
ISM represents a compressed document as symbols S , a dictionary D mapping symbols to short definitions, and relations G encoding precedence, exception, or application order. The main model is Qwen2.5-7B-Instruct in 4-bit inference with greedy decoding. Compression outputs are regenerated rather than truncated when they exceed budget. Synthetic Rule-QA provides hidden rule graphs for audit and oracle summaries.

RQ1: Dictionary Ablation

Condition	Acc	AR	CR
Full Context	0.750	1.000	1.000
Full Symbol + Dict	0.446	0.594	0.745
Deranged Dict	0.375	0.500	0.745
Flipped Dict	0.367	0.489	0.745
Symbol Only	0.413	0.550	0.079
Random Symbol	0.304	0.406	0.738

At $N=240$ paired questions, semantic dictionary flip is positive and significant ($\Delta_{\text{map_flip}}=+0.079$, $p=0.032$), and Symbol Only beats Random Symbol ($\Delta_{\text{symbol}}=+0.108$, $p=0.0005$). Derangement is not significant, indicating that surface label binding is not the main mechanism.

RQ1 contrast estimates



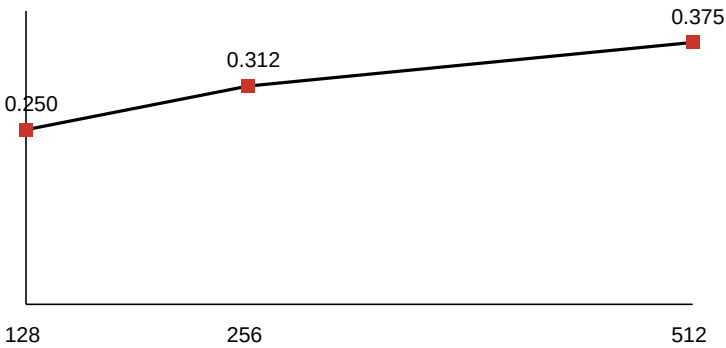
RQ3: Fixed-Budget Comparison

Method	128	256	512
Oracle Gold Summary	0.732 / 11.0	0.732 / 11.0	0.732 / 11.0
Model Summary	1.036 / 51.3	1.125 / 17.3	1.196 / 13.3
Keyword Extract	0.714 / 7.4	0.732 / 4.2	0.750 / 3.6
ISM	0.679 / 9.4	0.679 / 9.3	0.661 / 8.9

Budget	Summary-ISM	95% CI	McNemar p
128	+0.2500	[0.100, 0.400]	0.0029
256	+0.3125	[0.175, 0.450]	1.1e-4
512	+0.3750	[0.250, 0.500]	2.3e-7

The primary paired comparison is on the same 80 questions. Model Summary significantly outperforms ISM at every budget. Since both methods remove filler, this result is not explained by full-context degradation.

Summary accuracy advantage



RQ4: Reuse Cost-Accuracy

Method	n=1	n=8	n=32	n=64	Acc
Full Context	1354	10828	43312	86624	0.700
Model Summary	1526	2168	4369	7305	0.787
ISM	1548	2265	4725	8005	0.475
Oracle Gold	1530	2184	4428	7420	0.512
Keyword Extract	1820	3492	9225	16868	0.512

All cached methods become cheaper than full context from two questions onward. However, Model Summary is both cheaper and more accurate than ISM, so reuse is not an ISM-specific advantage. It is a property of any reusable text cache.

Discussion

What worked: ISM structure is used by the LLM. What failed: the original efficiency hypothesis. Prompt-only ISM is generated by an LLM and read by an LLM, just like a summary; within that class, ordinary summaries are stronger. The follow-up path is not another prompt format but a program-readable semantic IR with parser, checker, local transformations, and symbolic execution.

Limitations and Conclusion

The experiments are dev-scale and synthetic. QASPER and LLMLingua-2 are specified but not fully run in the reported results. Several GPU raw files are retained by hash and reproducibility commands rather than fully stored in the repository. Most importantly, this paper evaluates prompt-only ISM, not an executable semantic IR.

Conclusion: ISM is functionally used but is not a better prompt-only compressor than natural-language summaries. The honest contribution is a mixed result that maps the failure mode and points toward executable semantic IRs outside the LLM-only loop.

Selected References

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